# Spectrum and frame

A probabilistic view of the curvature that preconditioned optimizers estimate
Part I: What a power-law spectrum of a PSD matrix means

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**Abstract**

These notes build, from first principles and in stages, a probabilistic model of the curvature matrices that preconditioned optimizers such as SOAP estimate. The organising idea is to split a positive semi-definite matrix into its *spectrum*, a distribution of eigenvalues, and its *frame*, its eigenvectors. The lab's measurements show the first nearly stationary and the second rotating late in training. Part I treats one fixed matrix. It makes precise what a power-law spectrum is (four equivalent forms), what the exponent decides (which functionals are set by the top of the spectrum and which by the tail), how the spectrum becomes a probability distribution once the frame is unknown, and what a power law implies about the frame: eigenvalues are resolved far deeper than eigenvectors. It closes with what the literature already contains and with the questions that open Part II.

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## 1 Why these notes

Three results from the lab's CIFAR-10 studies (a depth-6 ViT; per-tensor generalized Gauss–Newton (GGN) blocks) motivate them.

(1) Late in training the per-tensor GGN spectrum is a nearly fixed power law with exponent $\alpha \approx 1$, while its eigenvectors rotate. A top-16 subspace cached 10 steps earlier keeps only 0.84–0.88 of the current top-16 curvature (`spectral_rotation_20261020`, [L]).

(2) A spectrum alone cannot precondition the bulk of the spectrum: a power-law diagonal reaches only the level of the exact diagonal. A few curvature products can (`bulk_krylov_20261014`, [L]).

(3) The best power of the curvature falls from about 3/4 early in training to 1/2 late, as the noise scales of strongly preconditioned steps grow (`noise_scale_20261021`, [L]).

SOAP already has the structure the first result suggests: an eigenbasis refreshed rarely (every 10 steps in our runs) and a diagonal second-moment estimate refreshed every step. The question behind these notes is whether SOAP was built from a model of how spectrum and frame evolve. If not, can such a model explain SOAP and improve it, for example by forecasting when to refresh the frame, how long to average, and how to treat momentum? Section 7 summarises what we found. SOAP's design is motivated by computational cost and by experiments, and we found no dynamical model of spectrum and frame behind it or elsewhere.

**Programme.**

**Part I** (this note): one PSD matrix. Spectrum and frame, power laws, the spectrum as a probability distribution, and what a power law implies about the frame.

**Part II** : dynamics. Matrices that move on or near the set of matrices with a fixed spectrum; drift and diffusion of the frame; models and their fingerprints in the lab's data.

**Part III** : inference. Priors with a known spectrum and an unknown frame; Krylov methods as inference on spectral measures; a stale frame with a fresh diagonal (EKFAC, SOAP) as an estimator.

**Part IV** : control. Forecasting refresh times, averaging windows and the treatment of momentum.

**Conventions.** Labels follow the lab's writing rules. *Propositions*, *lemmas* and *corollaries* are exact statements with proofs. *Approximations* are asymptotic or first-order statements with their conditions. *Observations* are measurements; they name the study and carry the evidence level [L] (local, at fixed states) or [T] (training). Numbers from synthetic computations are produced by `make_figures.py` in this folder; numbers from the lab's studies are taken from the studies' `RESULTS.md` files.

**Notation.** $\mathbb{S}^D$ is the set of real symmetric $D \times D$ matrices and $\mathbb{S}^D_+$ its positive semi-definite (PSD) subset; $\mathrm{O}(D)$ is the orthogonal group. Eigenvalues are sorted, $\lambda_1 \geq \cdots \geq \lambda_D$. $\|\cdot\|_{\mathrm{op}}$ is the operator norm and $\|\cdot\|_{\mathrm{F}}$ the Frobenius norm. $\delta_x$ is the point mass at $x$, $H_n = \sum_{k \leq n} 1/k$ the harmonic number, $\zeta$ the Riemann zeta function and $\gamma \approx 0.5772$ Euler's constant.

## 2 A PSD matrix is a spectrum and a frame

**Definition 2.1** (Spectrum and frame)**.** Every $A \in \mathbb{S}^D_+$ can be written $A = U\Lambda U^\top$ with $U \in \mathrm{O}(D)$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_D)$, $\lambda_1 \geq \cdots \geq \lambda_D \geq 0$. The vector $\sigma(A) = (\lambda_1, \ldots, \lambda_D)$ is the *spectrum* of $A$. Any such $U$ is a *frame* of $A$; its columns $u_k$ are eigenvectors.

**Proposition 2.2** (The orbit of a spectrum)**.** *Fix* $\Lambda$.

*(a) The set* $\mathcal{O}_\Lambda = \{Q\Lambda Q^\top : Q \in \mathrm{O}(D)\}$ *is the set of PSD matrices with spectrum* $\sigma = (\lambda_1, \ldots, \lambda_D)$.

(b) *If the $\lambda_k$ are distinct, the frame of $A \in \mathcal{O}_\Lambda$ is unique up to the signs of its columns, and $\mathcal{O}_\Lambda$ is a manifold of dimension $D(D-1)/2$.*

(c) *If the distinct values have multiplicities $m_1, \dots, m_r$, the frame is unique up to $\mathrm{O}(m_1) \times \cdots \times \mathrm{O}(m_r)$ acting within the eigenspaces, and $\dim \mathcal{O}_\Lambda = D(D-1)/2 - \sum_j m_j(m_j-1)/2$.*

*Proof.* (a) is the spectral theorem. The stabiliser $\{Q : Q\Lambda Q^\top = \Lambda\}$ consists of the $Q$ that commute with $\Lambda$, that is, that map every eigenspace to itself. These form $\mathrm{O}(m_1) \times \cdots \times \mathrm{O}(m_r)$, which is the finite group of diagonal sign matrices when all $m_j = 1$. The orbit is the homogeneous space $\mathrm{O}(D)$ modulo the stabiliser, of dimension $\dim \mathrm{O}(D) - \sum_j \dim \mathrm{O}(m_j) = D(D-1)/2 - \sum_j m_j(m_j-1)/2$. □

**Example 2.3** (Counting)**.** For the $64 \times 64$ weight blocks of the lab's ViT ($D = 4{,}096$) the spectrum is 4,096 numbers and the frame 8,386,560. For the $64 \times 256$ blocks ($D = 16{,}384$) they are 16,384 and 134,209,536. A power-law fit compresses the spectrum to two or three numbers. "The spectrum is stable" therefore constrains a vanishing part of the matrix; almost all of its information is in the frame.

**Definition 2.4** (Invariant and equivariant maps)**.** A function $F : \mathbb{S}^D \to \mathbb{R}$ is *invariant* if $F(QAQ^\top) = F(A)$ for all $Q \in \mathrm{O}(D)$. A map $G : \mathbb{S}^D \to \mathbb{S}^D$ is *equivariant* if $G(QAQ^\top) = Q\,G(A)\,Q^\top$ for all $Q \in \mathrm{O}(D)$.

**Proposition 2.5.** *(a) $F$ is invariant if and only if $F(A)$ depends on $A$ only through $\sigma(A)$.*

(b) *If $G$ is equivariant and $A$ has distinct eigenvalues, then $G(A) = \sum_k \varphi_k u_k u_k^\top$ for some numbers $\varphi_k$. In particular $G(A)$ commutes with $A$, and $G(A) = p(A)$ for a polynomial $p$ of degree less than $D$ (which may depend on $A$).*

*Proof.* (a) $A = U \operatorname{diag}(\sigma(A)) U^\top$, so invariance gives $F(A) = F(\operatorname{diag}\sigma(A))$; the converse is immediate. (b) The reflection $R_k = I - 2u_k u_k^\top$ satisfies $R_k A R_k^\top = A$, so equivariance gives $G(A) = R_k G(A) R_k$ and hence $R_k(G(A)u_k) = G(A)R_k u_k = -G(A)u_k$. The $-1$ eigenspace of $R_k$ is $\operatorname{span}(u_k)$, so $G(A)u_k = \varphi_k u_k$. Since the $\lambda_k$ are distinct, Lagrange interpolation gives $p$ with $p(\lambda_k) = \varphi_k$. □

*Remark* 2.6 (What this means for us)*.* Every statement about "the spectrum" (a power law, a trace, a condition number) is an invariant function. It takes the same value on the whole orbit and carries no information about the frame. Conversely, a preconditioner that must not depend on an unknown frame has to be equivariant, hence a function of the current matrix. Such a map is applied with matrix–vector products and never needs eigenvectors. Krylov methods (CG, Lanczos) are of this kind, equivariant jointly in the matrix and the start vector; a stored eigenbasis (SOAP, PARSEC-L) is not.

## 3 The spectral distribution, and four faces of a power law

**Definition 3.1** (Spectral distribution, counting function, rank function)**.** The *empirical spectral distribution* (ESD) of $A$ is $\mu_A = \frac{1}{D}\sum_{k=1}^{D} \delta_{\lambda_k}$. Its *counting function* is $N_A(t) = \#\{k : \lambda_k \geq t\} = D\,\mu_A([t, \infty))$, and the *rank function* is $k \mapsto \lambda_k$.

**Proposition 3.2** (A rank plot is a quantile plot)**.** *For $t \in \mathbb{R}$ and $k \in \{1, \dots, D\}$: $\lambda_k \geq t$ if and only if $N_A(t) \geq k$. Hence $\lambda_k = \max\{t : N_A(t) \geq k\}$: the rank function is the upper quantile function of $\mu_A$, $\lambda_k = Q_A(k/D)$ with $Q_A(p) = \max\{t : \mu_A([t, \infty)) \geq p\}$.*

*Proof.* The spectrum is sorted, so at least $k$ eigenvalues are $\geq t$ exactly when the $k$-th largest is. Thus $\{t : N_A(t) \geq k\} = (-\infty, \lambda_k]$. □

The log–log rank plots the lab fits are therefore log–log plots of the upper quantiles of $\mu_A$. They describe a distribution, read from its top.

**Definition 3.3** (Power law on a rank window)**.** *$A$ follows a power law with exponent $\alpha > 0$ and scale $c > 0$ on the window $\{1, \dots, K\}$, $K \le D$, if $\lambda_k = c\,k^{-\alpha}$ for $k \le K$. The law is pure if $K = D$. In data the statement is statistical: $\log \lambda_k = \log c - \alpha \log k + r_k$ with small residuals $r_k$, fitted on a chosen window.*

**Proposition 3.4** (Four faces of a power law)**.** *Let $\lambda_k = c\,k^{-\alpha}$ for $k \le K$, and $\lambda_{K+1} < \lambda_K$ if $K < D$. For every $t$ with $\lambda_K \le t \le c$:*

*(i)* rank: $\lambda_k = c\,k^{-\alpha}$ *for $k \le K$;*

*(ii)* counting: $N_A(t) = \lfloor (c/t)^{1/\alpha} \rfloor$*;*

*(iii)* distribution: $\mu_A([t, \infty)) = \lfloor (c/t)^{1/\alpha} \rfloor / D$*. Up to rounding this is the upper tail of a Pareto law with tail index $1/\alpha$. Smoothed, $\mu_A$ has density $\rho(t) = c^{1/\alpha} t^{-1-1/\alpha}/(\alpha D)$ on $[\lambda_K, c]$;*

*(iv)* depth: *with $v = \ln(c/t)$ the depth below the top eigenvalue (in nats), the number of eigenvalues within depth $v$ is $\lfloor e^{v/\alpha} \rfloor$ for $0 \le v \le \alpha \ln K$.*

*Conversely, (ii) on this range implies (i).*

*Proof.* For $t \ge \lambda_K$ no rank $k > K$ is counted, because $\lambda_k \le \lambda_{K+1} < \lambda_K \le t$. For $k \le K$, $c\,k^{-\alpha} \ge t$ if and only if $k \le (c/t)^{1/\alpha}$, and $(c/t)^{1/\alpha} \le K$. This gives (ii); (iii) divides by $D$ and differentiates, and (iv) substitutes $t = c\,e^{-v}$. Conversely, by Proposition 3.2, $\lambda_k = \max\{t : \lfloor (c/t)^{1/\alpha} \rfloor \ge k\} = c\,k^{-\alpha}$. □

*Remark* 3.5 (Reading the exponent). Doubling the rank lowers the eigenvalue by $\alpha$ octaves ($\lambda_{2k}/\lambda_k = 2^{-\alpha}$). Going one octave deeper multiplies the number of eigen-directions by $2^{1/\alpha}$; for $\alpha = 1$ each octave holds twice as many directions as the one above it. Most eigen-directions therefore sit near the bottom of the window (Figure 1c). A power law is a statement about a few large eigenvalues and very many small ones.

*Remark* 3.6 (Head law and global law). A fit on ranks $1, \dots, K$ with $K \ll D$ describes only the top $K/D$ of $\mu_A$. The lab fits ranks 1–200 in blocks with $D$ between 640 and 16,384, that is, the top 31% down to the top 1.2% of the eigen-directions. The rest of $\mu_A$, where most directions live, is not covered by the fit. Section 4 shows which quantities the fit nevertheless decides.

*Remark* 3.7 (Broken laws). An exponent $\alpha_1$ on ranks $k \le k_b$ and $\alpha_2$ on $k_b < k \le K$ (continuous at $k_b$) gives a piecewise power counting function: $\mu_A$ is a two-piece Pareto mixture.

*Observation* 3.8 (Broken early, straight late; [L]). Early in training the lab's whitened per-tensor GGN spectra are broken. Ranks 1–16 fall roughly as $k^{-1}$ and ranks 17–200 with $\alpha_{\text{bulk}} = 1.79$–$2.10$ at steps 10–25; the break flattens by step 100. A single law extrapolated from the top 16 then underestimates $\lambda_{17}$ by a factor 1.6–2.4 (`bulk_krylov_20261014`). The raw per-tensor GGN fits on ranks 1–200 go from $\alpha \approx 1.8$–$2.1$ at step 10 to $\alpha \approx 0.9$–$1.1$ at steps 250–500, with $R^2 \approx 0.99$ (`spectral_rotation_20261020`).

*Remark* 3.9 (Transformations of a law).

(a) $A \mapsto sA$ ($s > 0$) changes $c$ to $sc$ and keeps $\alpha$: the exponent is scale-free.

(b) $A \mapsto A^p$ ($p > 0$) gives exponent $\alpha p$ and scale $c^p$.

(c) In particular, preconditioning $A$ by a power of itself, $P = A^{-\nu}$ with $0 \le \nu < 1$, gives $P^{1/2} A P^{1/2} = A^{1-\nu}$: exponent $\alpha(1-\nu)$ and condition number $K^{\alpha(1-\nu)}$ on the window. The fractional-Newton family of `noise_scale_20261021` interpolates between the law ($\nu = 0$) and a flat spectrum ($\nu = 1$). Its best power late in training, $\nu \approx 1/2$, would halve the exponent of a power-law whitened spectrum.

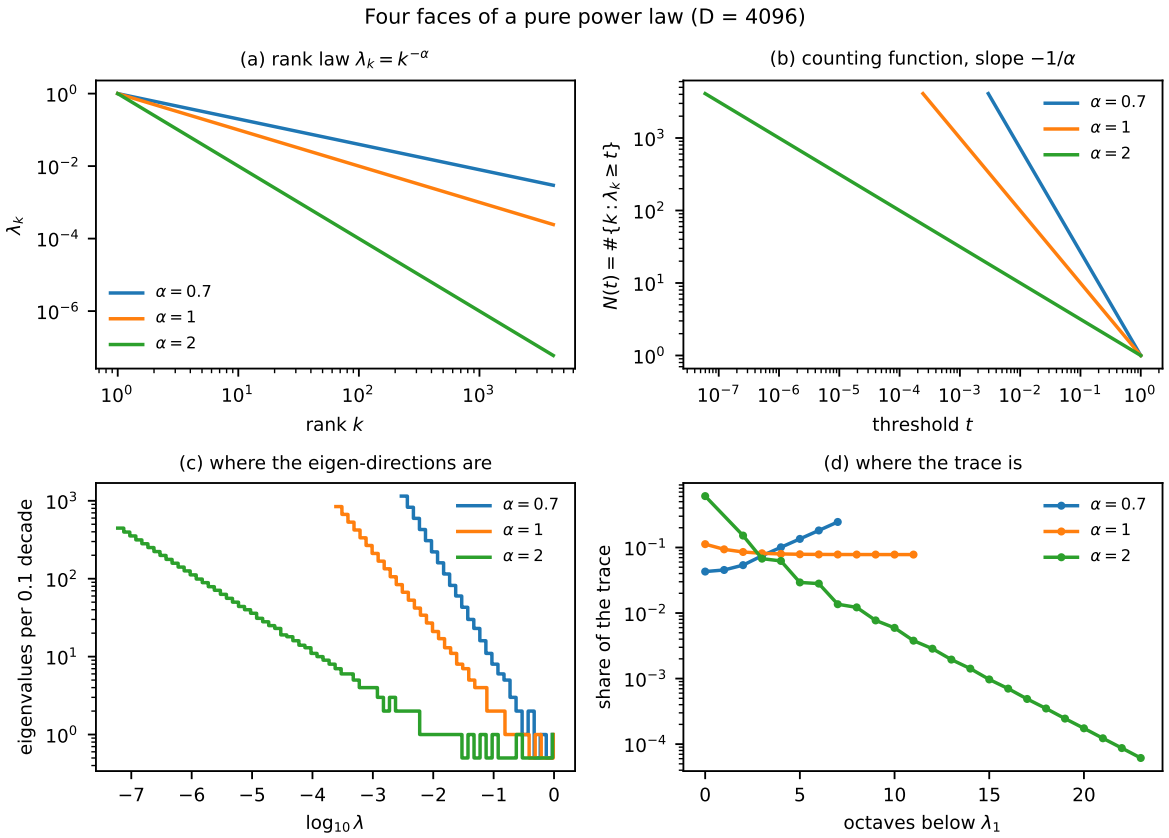



Figure 1: Four faces of a pure power law $\lambda_k = k^{-\alpha}$, $D = 4{,}096$ (synthetic). (a) The rank law. (b) Its counting function, with slope $-1/\alpha$ (Proposition 3.4). (c) Eigenvalues per 0.1 decade: most eigen-directions sit near the bottom (Remark 3.5). (d) Share of the trace per full octave below $\lambda_1$: top-heavy for $\alpha = 2$, flat for $\alpha = 1$, bottom-heavy for $\alpha = 0.7$ (Corollary 4.3).

(d) If $P$ is a function of a *different* matrix (another batch, an older step, SOAP's statistic), the spectrum of $P^{1/2}AP^{1/2}$ depends on the relative frame of the two matrices, not only on their spectra. For a uniformly random relative frame and large $D$, free probability describes the result (free multiplicative convolution). Part III returns to this.

*Remark* 3.10 (Kronecker products bend a law). Let $L \in \mathbb{S}^m$ and $R \in \mathbb{S}^n$ have pure laws $\ell_i = i^{-\alpha}$ and $r_j = j^{-\alpha}$. The eigenvalues of $R \otimes L$ are $(ij)^{-\alpha}$. For $x = t^{-1/\alpha} \leq \min(m, n)$,

$$N(t) = \#\{(i,j) : ij \leq x\} = \sum_{i \leq x} \lfloor x/i \rfloor = x \ln x + (2\gamma - 1)x + O(\sqrt{x})$$

(Dirichlet's divisor problem). Inverting, $\lambda_k \approx (k/\ln k)^{-\alpha}$: a power law up to a logarithm, with local log–log slope $-\alpha/(1 + 1/\ln x)$. For $m = 64$, $n = 256$ and $\alpha = 1$, a straight-line fit on ranks 17–200 gives exponent 0.77; the slope formula at the window's mean log rank gives 0.75. This matters because SOAP's frame is a Kronecker eigenbasis: exact laws in the factors appear shallower in the product.

## 4 What the exponent decides

**Lemma 4.1** (Power sums). *Let* $S_s(K) = \sum_{k=1}^{K} k^{-s}$.

*(a) For* $s > 1$*:* $S_s(K) \leq \zeta(s)$ *and* $\sum_{k>m} k^{-s} \leq m^{1-s}/(s-1)$*. The first* $m$ *ranks carry all but a fraction at most* $m^{1-s}/((s-1)\zeta(s))$*.*

(b) *For $s = 1$: $S_1(K) = H_K = \ln K + \gamma + O(1/K)$, and each dyadic block of ranks $\{2^j, \dots, 2^{j+1} - 1\}$ contributes $\ln 2 + O(2^{-j})$.*

(c) *For $0 \le s < 1$: $S_s(K) = K^{1-s}/(1-s) + O(1)$, and the ranks above $K/2$ carry a fraction $1 - 2^{s-1} + o(1)$. For $s < 0$: $S_s(K) = K^{1-s}/(1-s)\,(1 + o(1))$.*

*Proof.* For decreasing $f(x) = x^{-s}$ ($s \ge 0$), $\int_m^{M+1} f \le \sum_{k=m}^{M} f(k) \le f(m) + \int_m^M f$. Each claim follows by evaluating the integrals; for $s < 0$ the terms increase and the inequalities reverse. □

**Corollary 4.2** (Which part of a pure law decides a trace functional)**.** *For $\lambda_k = c\,k^{-\alpha}$, $k = 1, \dots, D$, we have $\operatorname{tr} A^q = c^q S_{\alpha q}(D)$, so*

| *Functional* | *decided by the top when* | *marginal at* | *decided by the tail when* |
|---|---|---|---|
| $\operatorname{tr} A^2 = \lVert A \rVert_{\mathrm{F}}^2$ | $\alpha > 1/2$ | $\alpha = 1/2$ | $\alpha < 1/2$ |
| $\operatorname{tr} A$ | $\alpha > 1$ | $\alpha = 1$ | $\alpha < 1$ |
| $\operatorname{tr} A^{1/2}$ | $\alpha > 2$ | $\alpha = 2$ | $\alpha < 2$ |
| $\operatorname{tr} A^{-1}$ | *never* | — | *always* ($\propto D^{1+\alpha}$) |

*A fit on the head decides only the top-dominated entries. Tail-dominated quantities depend on the part of $\mu_A$ the fit does not see: the trace for $\alpha < 1$, and every negative power, which is what preconditioning uses.*

**Corollary 4.3** (At $\alpha = 1$ every octave carries the same trace)**.** *For a pure law with $\alpha = 1$, every full octave of eigenvalues $[t, 2t)$ carries trace $c \ln 2 + O(t)$. The top-$k$ share of the trace is $H_k/H_D \approx \ln k / \ln D$. For the top 16 it is 0.380 at $D = 4{,}096$, 0.329 at $D = 16{,}384$ and 0.480 at $D = 640$.*

*Proof.* The eigenvalues in $[t, 2t)$ have ranks $c/(2t) < k \le c/t$, and $\sum c/k$ over these ranks is $c \ln 2 + O(t)$ by Lemma 4.1(b). The share is $\sum_{k \le 16}(c/k) / \sum_{k \le D}(c/k)$. □

*Observation* 4.4 (The late trace is consistent with an $\alpha \approx 1$ law; [L])*.* In `spectral_rotation_20261020` (raw per-tensor GGN on one fixed batch), the late fits (steps 250–500) give $\alpha \approx 0.9$–$1.1$ on ranks 1–200, with a slightly flatter head ($\alpha_{\mathrm{top}} = 0.65$–$0.86$) and $\alpha_{\mathrm{bulk}} = 0.96$–$1.17$. The median top-16 share of the exact block trace over the 39 tensors is 0.35–0.45 at step 400 and 0.33–0.42 at step 500. A pure $\alpha = 1$ law over all ranks gives 0.380 for the 24 blocks with $D = 4{,}096$ and 0.329 for the 12 with $D = 16{,}384$. The trace is the only measured quantity that sees the whole spectrum, and it is consistent with the law extending well below rank 200. It is one number per tensor, so this is a consistency check, not a test.

*Observation* 4.5 (Spectra leave the law at the bottom; [L])*.* In `noise_scale_20261021` the smallest Ritz value of a damped whitened block $K + \delta I$ was $0.9987\,\delta$, so at least one block has eigenvalues at numerical zero. Measured spectra leave any power law at the bottom (finite data, symmetries). The tail-dominated rows of Corollary 4.2 are then set by the damping, not by the law.

**Proposition 4.6** (Effective dimension under damping)**.** *For $\delta > 0$ let*

$$d_\delta(A) = \operatorname{tr} A(A + \delta I)^{-1} = \sum_k \frac{\lambda_k}{\lambda_k + \delta}.$$

(a) *For every $A \in \mathbb{S}_+^D$: $\frac{1}{2} N_A(\delta) \le d_\delta(A) \le N_A(\delta) + \delta^{-1} \sum_{\lambda_k < \delta} \lambda_k$.*

(b) *For a pure law with $\alpha > 1$ and $D = \infty$: $\left| d_\delta - (c/\delta)^{1/\alpha} \frac{\pi/\alpha}{\sin(\pi/\alpha)} \right| \le 1$.*

(c) *For a pure law with $\alpha = 1$: $\left| d_\delta - (c/\delta) \ln(1 + D\delta/c) \right| \le 1$.*

*Proof.* (a) Terms with $\lambda_k \geq \delta$ lie in $[1/2, 1]$; terms with $\lambda_k < \delta$ are at most $\lambda_k/\delta$. (b) $f(x) = 1/(1+(\delta/c)x^\alpha)$ is decreasing with $f(0) = 1$, so $|\sum_{k\geq 1} f(k) - \int_0^\infty f| \leq 1$, and $\int_0^\infty dx/(1+ax^\alpha) = a^{-1/\alpha}\int_0^\infty du/(1+u^\alpha) = a^{-1/\alpha}(\pi/\alpha)/\sin(\pi/\alpha)$. (c) The same comparison with $\int_0^D dx/(1+\delta x/c)$. □

Numerical check: for $\alpha = 2$ and $\delta = 10^{-6}c$ the sum (to rank $10^7$) is 1570.2 against 1570.8; for $\alpha = 1$, $D = 4{,}096$ and $\delta = 10^{-3}c$ it is 1628.1 against 1628.5. Damping $\delta$ effectively truncates the law at rank $N_A(\delta) = (c/\delta)^{1/\alpha}$. At $\alpha = 1$ the directions below $\delta$ still add a logarithmic factor, so the same damping keeps very different numbers of directions active for different exponents.

# 5 The spectrum as a probability distribution

## 5.1 Spectrum known, frame unknown

**Definition 5.1** (The Haar model)**.** Fix $\Lambda$, draw $Q$ from the Haar (uniform) distribution on $\mathrm{O}(D)$, and set $A = Q\Lambda Q^\top$. The law of $A$ is the unique probability distribution on $\mathcal{O}_\Lambda$ that is invariant under all rotations $A \mapsto RAR^\top$. It models a matrix whose spectrum is known and whose frame is not.

**Proposition 5.2** (The eigenvalue an unmeasured direction meets)**.** *In the Haar model, let $v$ be a fixed unit vector and $w_k = (q_k^\top v)^2$, with $q_k$ the columns of $Q$.*

*(a)* $(w_1, \ldots, w_D) \sim \mathrm{Dirichlet}(\frac{1}{2}, \ldots, \frac{1}{2})$*; in particular* $\mathbb{E}w_k = 1/D$*.*

*(b) Let $I$ be a random index with* $\Pr(I = k \mid Q) = w_k$*. Then* $\Pr(I = k) = 1/D$*, so* $\lambda_I \sim \mu_A$*.*

*(c) For every $f$, $v^\top f(A)v = \sum_k f(\lambda_k)w_k = \mathbb{E}[f(\lambda_I) \mid Q]$, and*

$$\mathbb{E}[v^\top f(A)v] = \int f\, d\mu_A, \qquad \mathrm{Var}[v^\top f(A)v] = \frac{2\,\mathrm{Var}_{\mu_A}(f)}{D+2}.$$

*Proof.* $Q^\top v$ is uniform on the unit sphere because $Q$ is Haar distributed. A uniform unit vector is $g/\|g\|$ with $g \sim \mathcal{N}(0, I_D)$. The $g_k^2$ are independent $\mathrm{Gamma}(\frac{1}{2}, 2)$ variables, and independent Gamma variables with a common scale, normalised by their sum, are Dirichlet distributed: this gives (a), and (b) follows from $\mathbb{E}w_k = 1/D$. For (c), the Dirichlet moments are $\mathrm{Var}\, w_k = 2(D-1)/(D^2(D+2))$ and $\mathrm{Cov}(w_k, w_l) = -2/(D^2(D+2))$, so $\mathrm{Var}\sum_k f_k w_k = \frac{2}{D^2(D+2)}(D\sum_k f_k^2 - (\sum_k f_k)^2) = 2\,\mathrm{Var}_{\mu_A}(f)/(D+2)$. □

**Example 5.3.** For a pure $\alpha = 1$ law with $D = 4{,}096$, the Rayleigh quotient $v^\top Av$ of a uniformly random direction has mean $\mathrm{tr}\, A/D$ and relative standard deviation 0.203. A Monte Carlo check (200,000 draws, $D = 256$) reproduces the variance formula with ratio 1.0016 and the mean with ratio 0.9995.

*Remark* 5.4 (The probabilistic-numerics reading). Proposition 5.2 is the precise sense in which "the spectrum is a distribution". When nothing is known about the frame, $\mu_A$ is the prior predictive distribution of the curvature that any direction will meet. A matrix–vector product $Av$ is an observation that changes this: it reveals part of the spectral measure seen from $v$ (Section 5.2). In the language of probabilistic numerics (Hennig 2015; Wenger & Hennig 2020), the spectrum plays the role of the prior, and computation, the products, is inference about the frame relative to the vectors we care about. Part III develops this.

## 5.2 The spectral measure seen from a vector

**Definition 5.5.** For $v \in \mathbb{R}^D$, the spectral measure of $A$ seen from $v$ is $\mu_{A,v} = \sum_k (u_k^\top v)^2 \, \delta_{\lambda_k}$, of total mass $\|v\|^2$; its normalisation is $\nu_{A,v} = \mu_{A,v}/\|v\|^2$.

**Proposition 5.6.** *(a)* $\int f \, d\mu_{A,v} = v^\top f(A) v$ *for every* $f$.

*(b)* $\mu_{QAQ^\top, Qv} = \mu_{A,v}$ *for all* $Q \in \mathrm{O}(D)$.

*(c) In the Haar model,* $\mathbb{E}\,\mu_{A,v} = \|v\|^2 \mu_A$ *as measures.*

*(d) (Golub & Meurant 2010) Lanczos run for* $m$ *steps on* $A$ *from* $v$ *returns nodes* $\theta_1, \dots, \theta_m$ *and weights* $\tau_1, \dots, \tau_m$ *(the squared first components of the eigenvectors of its tridiagonal matrix) with* $\sum_j \tau_j \theta_j^i = \int \lambda^i \, d\nu_{A,v}(\lambda)$ *for* $0 \le i \le 2m-1$*: it computes the* $m$*-point Gauss quadrature rule of* $\nu_{A,v}$.

*(e) For* $A \succ 0$*, conjugate gradients on* $Ax = b$ *from* $x_0 = 0$ *satisfies*

$$\|x - x_m\|_A^2 = \min\Big\{ \int p(\lambda)^2 \lambda^{-1} \, d\mu_{A,b}(\lambda) : \deg p \le m, \; p(0) = 1 \Big\}.$$

*Proof.* (a) $A = \sum_k \lambda_k u_k u_k^\top$. (b) $QAQ^\top$ has eigenvectors $Qu_k$, and $(Qu_k)^\top Qv = u_k^\top v$. (c) Proposition 5.2(a). (d) Golub & Meurant (2010). (e) CG's iterate $x_m = q(A)b$ minimises the $A$-norm error over polynomials $q$ of degree less than $m$. With $p(\lambda) = 1 - \lambda q(\lambda)$, $x - x_m = p(A)A^{-1}b$ and $\|x - x_m\|_A^2 = b^\top p(A)^2 A^{-1} b = \int p^2 \lambda^{-1} \, d\mu_{A,b}$. □

So the frame enters Krylov methods only through the weights of the start vector. Two matrices with different frames but the same measure seen from $b$ produce the same CG errors. (The `weights` in the lab's `noise_scale_20261021/scripts/lanczos.py` are exactly the $\tau_j$ of (d).)

**Proposition 5.7** (One step on a quadratic)**.** *Let* $A \succ 0$, $g \in \mathbb{R}^D$ *and* $q(x) = \tfrac{1}{2} x^\top A x - g^\top x$. *For* $f > 0$ *on the spectrum, the best decrease along* $x = \alpha f(A) g$ *is*

$$\max_\alpha [q(0) - q(\alpha f(A) g)] = \frac{\left( \int f \, d\mu_{A,g} \right)^2}{2 \int \lambda f(\lambda)^2 \, d\mu_{A,g}(\lambda)}.$$

*Proof.* $q(\alpha f(A) g) = \tfrac{1}{2} \alpha^2 g^\top f(A) A f(A) g - \alpha g^\top f(A) g$. Maximise over $\alpha$ and use Proposition 5.6(a). □

**Corollary 5.8** (The value of curvature is a spread)**.** *For Newton (*$f = 1/\lambda$*) against a tuned gradient step (*$f = 1$*), the ratio of best decreases is* $\mathbb{E}_\nu[\lambda]\, \mathbb{E}_\nu[\lambda^{-1}]$ *with* $\nu = \nu_{A,g}$. *It lies in* $[1, (\kappa+1)^2/(4\kappa)]$, *with* $\kappa$ *the condition number on the support of* $\nu$ *(Kantorovich). For a vector with uniform spectral weights (*$\nu = \mu_A$*) and a pure* $\alpha = 1$ *law it equals* $H_D(D+1)/(2D) \approx \tfrac{1}{2} \ln D$*: 4.45 at* $D = 4{,}096$ *and 5.14 at* $D = 16{,}384$.

*Proof.* Proposition 5.7 gives

$$\text{Newton: } \tfrac{1}{2} \int \lambda^{-1} \, d\mu_{A,g}, \qquad \text{gradient: } \frac{\left( \int d\mu_{A,g} \right)^2}{2 \int \lambda \, d\mu_{A,g}};$$

divide. For the pure law, $\mathbb{E}_\mu[\lambda] = c H_D / D$ and $\mathbb{E}_\mu[\lambda^{-1}] = (D+1)/(2c)$. □

*Remark* 5.9 (Two measures decide a step). With batch noise of covariance $\Sigma$, the denominator of Proposition 5.7 gains the term $(1/B - 1/N) \int \lambda f^2 \, d\mu_{A,\Sigma}$, where $\mu_{A,\Sigma} = \sum_k (u_k^\top \Sigma u_k) \, \delta_{\lambda_k}$ (`batch_noise.md`, T1). For one step, all that matters about $A$ is the pair $(\mu_{A,\bar{g}}, \mu_{A,\Sigma})$: how the signal and the noise are spread over the spectrum. The frame enters only through these weights.

Two questions follow for training. Does the gradient's measure $\nu_{A,\bar{g}}$ stay stationary while the frame rotates, that is, does the gradient co-rotate with the frame? And how do it and $\mu_{A,\Sigma}$ change during training? The forms $z^\top A^q z$, $q \in [-1, 1]$, measured in `noise_scale_20261021` are moments of exactly these measures.

## 6 What a power law implies about the frame

**Proposition 6.1** (Eigenvalues are stable)**.** *For $A, B \in \mathbb{S}^D$ with sorted eigenvalues:*

*(a)* $|\lambda_k(A) - \lambda_k(B)| \le \|A - B\|_{\mathrm{op}}$ *for every $k$ (Weyl);*

*(b)* $\sum_k (\lambda_k(A) - \lambda_k(B))^2 \le \|A - B\|_{\mathrm{F}}^2$ *(Hoffman–Wielandt);*

*(c) hence* $W_2(\mu_A, \mu_B) \le \|A - B\|_{\mathrm{F}}/\sqrt{D}$.

*Proof.* (a) and (b): Bhatia (1997). (c) On the real line the monotone coupling is optimal for the Wasserstein-2 distance, so $W_2(\mu_A, \mu_B)^2 = D^{-1} \sum_k (\lambda_k(A) - \lambda_k(B))^2$. □

**Proposition 6.2** (Eigenvectors are only as stable as their gaps)**.** *(Davis–Kahan, in the form of Yu, Wang & Samworth 2015.) Let $A, B \in \mathbb{S}^D$, $E = B - A$, and $g_k = \min(\lambda_{k-1}(A) - \lambda_k(A), \lambda_k(A) - \lambda_{k+1}(A)) > 0$ with $\lambda_0 = +\infty$, $\lambda_{D+1} = -\infty$. Then $\sin \angle(u_k(A), u_k(B)) \le 2\|E\|_{\mathrm{op}}/g_k$. For the top-$k$ subspaces, $\|\sin \Theta\|_{\mathrm{F}} \le 2 \min(\sqrt{k}\|E\|_{\mathrm{op}}, \|E\|_{\mathrm{F}})/(\lambda_k(A) - \lambda_{k+1}(A))$.*

**Lemma 6.3** (Gaps of a power law)**.** *If $\lambda_k = c\,k^{-\alpha}$, then $\frac{\alpha}{k}(1 + \frac{1}{k})^{-\alpha-1} \le (\lambda_k - \lambda_{k+1})/\lambda_k \le \frac{\alpha}{k}$, and for $2 \le k < D$ the gap $g_k$ of Proposition 6.2 equals $\lambda_k - \lambda_{k+1}$.*

*Proof.* $h(x) = 1 - (1 + x)^{-\alpha}$ is concave with $h(0) = 0$ and $h'(x) = \alpha(1 + x)^{-\alpha-1}$, so $xh'(x) \le h(x) \le xh'(0)$; take $x = 1/k$. The map $k \mapsto c\,k^{-\alpha}$ is convex and decreasing, so consecutive gaps decrease and the gap below is the smaller one. □

**Proposition 6.4** (Two resolution limits)**.** *Let $A$ follow a pure law $(c, \alpha)$ and $\|B - A\|_{\mathrm{op}} \le \eta$. For $s \in (0, 1]$:*

*(a)* $|\lambda_k(B) - \lambda_k(A)| \le s\lambda_k(A)$ *for every $k \le K_{val} := (sc/\eta)^{1/\alpha}$;*

*(b)* $\sin \angle(u_k(A), u_k(B)) \le s$ *for every $k < D$ with $k \le K_{vec} := \left(s\alpha c/(2^{\alpha+2}\eta)\right)^{1/(\alpha+1)}$;*

*(c) $K_{vec}/K_{val} \to 0$ as $\eta \to 0$: for small perturbations far more eigenvalues are resolved than eigenvectors.*

*Proof.* (a) Weyl: $|\Delta\lambda_k| \le \eta \le sc\,k^{-\alpha}$ if and only if $k \le (sc/\eta)^{1/\alpha}$. (b) By Lemma 6.3 and $1 + 1/k \le 2$, $g_k \ge \alpha c\,k^{-\alpha-1}2^{-\alpha-1}$ (for $k = 1$ the gap is $\lambda_1 - \lambda_2$, covered by the same bound). Proposition 6.2 gives $\sin \le 2\eta/g_k \le 2^{\alpha+2}\eta k^{\alpha+1}/(\alpha c) \le s$ for $k \le K_{\mathrm{vec}}$. (c) $K_{\mathrm{vec}}/K_{\mathrm{val}} \propto \eta^{1/\alpha - 1/(\alpha+1)} = \eta^{1/(\alpha(\alpha+1))} \to 0$. □

These are worst-case guarantees. Typical perturbations behave alike, with explicit constants.

**Approximation 6.5** (Typical perturbations)**.** Let $E$ have independent centred Gaussian entries, with variance $\varepsilon^2$ off the diagonal and $2\varepsilon^2$ on it, so that $\|E\|_{\mathrm{op}} \approx 2\varepsilon\sqrt{D}$. While the result is small, first-order perturbation theory gives

$$\sin^2 \angle(u_k(A), u_k(B)) \approx \varepsilon^2 \sum_{j \ne k} (\lambda_k - \lambda_j)^{-2}.$$

For a pure law the sum has two parts. The neighbouring ranks contribute $\sum_{m \ne 0} (mg_k)^{-2} \approx 2\zeta(2)k^{2\alpha+2}/(\alpha^2 c^2)$, and the many much smaller eigenvalues about $Dk^{2\alpha}/c^2$. The neighbours

dominate for $k \gtrsim \alpha\sqrt{D/(2\zeta(2))}$. The typical eigenvector resolution limit, where the sine reaches $s$, is then

$$k_{\mathrm{vec}} \approx \Big(\frac{s\alpha c}{\sqrt{2\zeta(2)}\,\varepsilon}\Big)^{1/(\alpha+1)}.$$

An isolated eigenvalue moves by about $\sqrt{2}\varepsilon$ (its diagonal entry), and no eigenvalue moves by more than $\|E\|_{\mathrm{op}}$.

Figure 2 checks this for a pure $\alpha = 1$ law with $D = 2{,}000$ and $\varepsilon = 10^{-5}$ ($\|E\|_{\mathrm{op}} \approx 8.9 \times 10^{-4}$; medians of three draws).

- *Eigenvalues:* relative errors stay below $3.4 \times 10^{-3}$ in the top 100 ranks and reach 0.5 only at rank 1,801 (Weyl's worst case: rank 560).
- *Eigenvectors:* the first-order formula matches the simulated angles, and the sine reaches 0.5 at rank 169 (predicted 166).

Eigenvalues are resolved about ten times deeper than eigenvectors here.

**Proposition 6.6** (Lipschitz spectral functions are as stable as the matrix)**.** *If $f$ is $L$-Lipschitz on an interval containing the spectra of $A, B \in \mathbb{S}^D$, then $\|f(A) - f(B)\|_{\mathrm{F}} \le L\|A - B\|_{\mathrm{F}}$.*

*Proof.* Write $A = U_A\Lambda_A U_A^\top$, $B = U_B\Lambda_B U_B^\top$ and $W = U_A^\top U_B$. Then $U_A^\top(A - B)U_B = \Lambda_A W - W\Lambda_B$ has entries $(a_i - b_j)W_{ij}$, and $U_A^\top(f(A) - f(B))U_B$ has entries $(f(a_i) - f(b_j))W_{ij}$. Since $|f(a_i) - f(b_j)| \le L|a_i - b_j|$ entrywise and the Frobenius norm is invariant under orthogonal factors, the claim follows. □

**Corollary 6.7** (Damping sets the stability of a fractional step)**.** *$f(\lambda) = (\lambda + \delta)^{-\nu}$ is $\nu\delta^{-\nu-1}$-Lipschitz on $[0, \infty)$, so $\|(A+\delta I)^{-\nu} - (B+\delta I)^{-\nu}\|_{\mathrm{F}} \le \nu\delta^{-\nu-1}\|A - B\|_{\mathrm{F}}$, whatever the eigenvectors did. The top-$k$ projector $\Pi_k$ has no bound of this kind. By Proposition 6.2, $\|\Pi_k(A) - \Pi_k(B)\|_{\mathrm{F}} = \sqrt{2}\|\sin\Theta\|_{\mathrm{F}} \le 2\sqrt{2}\min(\sqrt{k}\|E\|_{\mathrm{op}}, \|E\|_{\mathrm{F}})/(\lambda_k - \lambda_{k+1})$, and for a power law the gap at the cut is about $\alpha\lambda_k/k$ (Lemma 6.3).*

*Remark* 6.8 (A cut or a smooth function?). The two bounds have different strengths, and Figure 2c shows both effects. A hard cut where the gap is large is far more stable than a smooth "soft cut" $A(A+\lambda_k I)^{-1}$ at the same place: at $k = 16$ the relative errors are 0.007 (hard) and 0.0739 (soft). The smooth function acts on every direction and pays for the perturbation everywhere; its error is controlled by $\|E\|_{\mathrm{F}}$, which grows like $\varepsilon D$. Deep in the spectrum, where gaps are tiny, the order reverses: at $k \approx 1{,}000$ the errors are 0.54 (hard) and 0.214 (soft). The curves cross at $k \approx 346$, about twice the eigenvector resolution limit. (Soft-cut values are on a grid of cut positions; the ones quoted are at 15, 193 and 917.)

The design rule this suggests is not "never store a frame" but: cut only where the frame is resolved, and use smooth functions of the current matrix beyond that. PARSEC-L's brake $I - \sum_{k \le r} \frac{\kappa_k}{1+\kappa_k} w_k w_k^\top$ with $\kappa_k = c_{\mathrm{b}}\eta\lambda_k$ is exactly the soft function $\lambda/(\lambda + 1/(c_{\mathrm{b}}\eta))$ applied inside a hard top-$r$ cut. Its fully smooth version is $(I + c_{\mathrm{b}}\eta K)^{-1}$, which needs no eigenvectors.

**Proposition 6.9** (Stale frame, fresh diagonal)**.** *Let $U \in \mathrm{O}(D)$ be any frame, for example the eigenframe of an earlier matrix, and let $A \in \mathbb{S}^D$. Define the pinching $\mathcal{P}_U(A) = U\,\mathrm{diag}(u_1^\top A u_1, \dots, u_D^\top A u_D)\,U^\top$.*

*(a) $\mathcal{P}_U(A)$ is the closest matrix to $A$ in Frobenius norm among matrices with frame $U$, and $\|A - \mathcal{P}_U(A)\|_{\mathrm{F}}^2 = \sum_{i \ne j}(u_i^\top A u_j)^2$.*

*(b) $\operatorname{tr}\mathcal{P}_U(A) = \operatorname{tr} A$, and the eigenvalues of $\mathcal{P}_U(A)$, the Rayleigh quotients $u_i^\top A u_i$, are majorised by those of $A$. For every $k$, the sum of the $k$ largest Rayleigh quotients is at most $\lambda_1(A) + \cdots + \lambda_k(A)$. In particular $\max_i u_i^\top A u_i \le \lambda_1(A)$ and $\min_i u_i^\top A u_i \ge \lambda_D(A)$.*

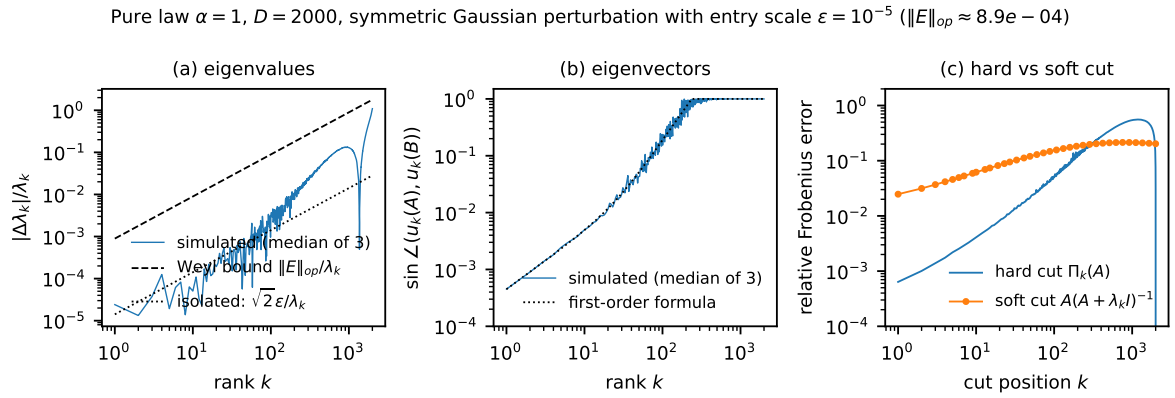



Figure 2: Resolution limits (synthetic). A pure $\alpha = 1$ law, $D = 2{,}000$, perturbed by a symmetric Gaussian matrix with entry scale $\varepsilon = 10^{-5}$; medians of three draws. (a) Relative eigenvalue errors, with Weyl's bound and the isolated-eigenvalue estimate $\sqrt{2}\varepsilon/\lambda_k$. (b) Eigenvector angles, with the first-order formula of Approximation 6.5. (c) Relative Frobenius error of the top-$k$ projector (hard cut) and of the soft cut $A(A + \lambda_k I)^{-1}$ at the same place (Remark 6.8).

*Proof.* (a) $U^\top(A - UMU^\top)U = U^\top AU - M$ for diagonal $M$: the off-diagonal part is fixed, and the diagonal part vanishes for $M = \mathrm{diag}(U^\top AU)$. (b) Schur's theorem: the diagonal of a symmetric matrix is majorised by its eigenvalues (Horn & Johnson 2013, Section 4.3). □

*Remark* 6.10 (The bridge to SOAP). This is the structure of EKFAC and SOAP: a frame refreshed rarely and a diagonal refreshed every step. Part (a) is the Frobenius-optimality argument that George et al. (2018) give for EKFAC's diagonal, and Eschenhagen et al. (2025) analyse Shampoo and SOAP in the same Frobenius terms. Part (b) adds what a stale frame costs spectrally: the effective spectrum is flattened, with the top lowered and the bottom raised, the more so the further the frame has drifted. That is the direction of the shrinkage that rotationally invariant estimators from random-matrix statistics apply when eigenvectors are unreliable (Bun, Bouchaud & Potters 2017). Two cautions for Part III: SOAP's diagonal is an exponential average of squared rotated gradients, a second moment of the gradient rather than the GGN; and it enters through a square root.

*Observation* 6.11 (Late rotation is not only near-tie mixing; [L]). Late in training, the 10-step capture of a cached top-$k$ subspace rises with $k$: 0.78–0.84 at $k = 1$ and 0.87–0.91 at $k = 200$ (`spectral_rotation_20261020`). The study read this as mixing within nearly tied groups, which is what Proposition 6.4 and Approximation 6.5 predict for a power law. But the late head is flatter ($\alpha_{\mathrm{top}} = 0.65$–$0.86$), so the relative gap below the top eigenvalue is $1 - 2^{-\alpha_{\mathrm{top}}} \approx 0.36$–$0.45$, which is not small. Losing 16–22% of the top direction's curvature in 10 steps therefore means the 10-step change is not small compared with the top gap. Part II has to allow changes that are large at the top, not only mixing among near-ties. Rotation also exceeds batch noise: at fixed weights another batch's top-16 subspace captures 0.93–0.999.

# 7 What is known, and what is not (searches of 23 October 2026)

The motivating question was whether SOAP was built from a model of how spectrum and frame evolve. References marked * were checked online on 23 October 2026, at least at the level of the abstract; the others are cited from memory.

- **EKFAC** (George et al. 2018*). The eigenbasis is expensive and is recomputed rarely; the

diagonal scaling in that basis is cheap and tracked every step. The justification is cost plus a Frobenius-norm argument (a provably better approximation of the Fisher than K-FAC). There is no model of how the eigenbasis moves.

- **SOAP** (Vyas et al. 2024*). Shampoo with power 1/2 equals Adafactor in Shampoo's eigenbasis, and SOAP runs Adam there instead. The eigenbasis is described as slowly changing, and the second moments as cheap enough to update every step. The refresh interval (1–100) is studied empirically: SOAP degrades more slowly with it than Shampoo does. There is no model of eigenbasis drift against eigenvalue drift and no theory of the optimal interval. Momentum is kept in parameter space and projected into the current basis at every step; the second-moment estimate lives in the rotated space and, on our reading of their Algorithm 3, is not re-rotated when the basis changes (to be verified). Vyas et al. note that Anil et al. (2020, Appendix B) described an essentially equivalent algorithm.

- **Purifying Shampoo** (Eschenhagen et al. 2025*). A Frobenius-norm view of Shampoo's heuristics. It decouples the updates of eigenvalues and eigenbasis, shows that correcting the eigenvalues replaces learning-rate grafting, and proposes an adaptive criterion for recomputing the eigenbasis, based on the termination rule of a warm-started QR iteration. This is the closest existing work to adaptive refreshing. Its criterion controls the current approximation error; it does not forecast the error from a model of the frame's motion.

- **Power-law Hessian spectra.** Xie et al. (2022*) report rank laws $\lambda_i \propto i^{-s}$ with $s \approx 1$ in well-trained networks and give a maximum-entropy explanation; Tang et al. (2025*) extend this to CNNs and LLMs. Meterez et al. (2026), the owner's reference for the "stable law, rotating eigenvectors" picture, was not found by our searches and is unverified here.

- **Eigenvector dynamics in training.** Gur-Ari, Roberts & Dyer (2018) find gradients concentrated in the top Hessian subspace. Marjankowska, Modugno & Barucca (2026*) track the displacement and localisation of leading Hessian eigenvectors under SGD and Adam; they do not connect it to spectral stability or to SOAP.

- **Random-matrix statistics.** Rotationally invariant estimators keep the sample eigenvectors and replace the eigenvalues by a nonlinear shrinkage computed from the spectrum (Ledoit & Péché 2011; Bun et al. 2016*; Bun, Bouchaud & Potters 2017*). Allez & Bouchaud (2012) study eigenvector dynamics of exponentially weighted covariance matrices, the estimator family of SOAP's statistics. We found no connection to SOAP or Shampoo.

- **Probabilistic numerics.** Gaussian inference on Hessians (Hennig & Kiefel 2013; Hennig 2015), active inference of a preconditioner from noisy Hessian-vector products (de Roos & Hennig 2019*), calibrated probabilistic linear solvers (Wenger & Hennig 2020), and the bias of mini-batch quadratics (Tatzel, Mucsányi, Hackel & Hennig 2025*).

**Conclusion.** As far as these searches show, no published work treats the spectral law as (nearly) stationary and the frame as a latent process, and derives SOAP's refresh schedule, averaging windows or treatment of momentum from such a model. Searches can miss work; this should be re-checked before any claim of novelty.

# 8 Where Part II starts

**Plan for Part II (dynamics).**

(1) *Identity:* split a change of $A$ into a spectral part and a rotation. With $W = U^\top \dot{U}$ skew-symmetric, $U^\top \dot{A} U = \dot{\Lambda} + W\Lambda - \Lambda W$: the diagonal holds $\dot{\lambda}_k$ and the off-diagonal $(\lambda_j - \lambda_i)W_{ij}$. "Stable spectrum" means the change is nearly all off-diagonal.

(2) *Three stochastic pictures with different fingerprints.*

- Isotropic matrix noise (Dyson Brownian motion): eigenvector pairs diffuse at rates $\propto (\lambda_i - \lambda_j)^{-2}$.
- Frame diffusion (Brownian motion on $\mathrm{O}(D)$): rates independent of the gaps.
- Deterministic drift: ballistic, hence predictable.

Their fingerprints are capture against lag (linear or quadratic onset), rotation against gap, and replicas forked with different data orders.

(3) *Structure:* the change of a GGN under a weight step is not isotropic; what is its structure?

(4) *Data:* `spectral_rotation_20261020` saved spectra and captures. Replays (10–12 minutes per trajectory) give eigenvectors. `noise_scale_20261021`'s forms give the moments of the gradient's and the noise's spectral measures at 36 states without new compute.

**Questions for us.**

*Question* 1. Which matrix should the theory treat first: the GGN, which we measured, or SOAP's own statistic, Kronecker factors of gradient second moments (Remark 3.10)?

*Question* 2. Is a scale-free law the right null hypothesis, given that the trace grows 1.3–1.6× during the learning-rate decay while the shape does not (`spectral_rotation_20261020`)?

*Question* 3. Which prior on the frame: uniform (Haar), or concentrated around SOAP's current basis (a matrix Bingham or von Mises–Fisher law)?

*Question* 4. Which first measurement: rotation against gap from replays, or the gradient's spectral measure, which the forms saved by `noise_scale_20261021` give for free?

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